LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	33
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose and explanation of electro pneumatic servo actuators and electro mechanical servo actuators
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose and explanation of electro pneumatic servo actuators and electro mechanical servo actuators.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagrams and signal flow charts from IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Define a servo actuator in the context of flight controls.
(c) List two types of servo actuators covered in this lesson.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of electro pneumatic servo actuators • Operating principle of electro pneumatic servo actuators • Key components of electro pneumatic servo actuators • Advantages and limitations of electro pneumatic systems	L & Ds	PPT, IB	15 min
(b)	• Definition and purpose of electro mechanical servo actuators • Operating principle of electro mechanical servo actuators • Key components of electro mechanical servo actuators • Comparison between electro pneumatic and electro mechanical types	L & Ds	PPT, PDF	20 min
(c)	• Role of servo actuators in AFCS control loops • Signal conversion processes in electro pneumatic and electro mechanical systems • Maintenance considerations for servo actuators • Troubleshooting common servo actuator failures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how an electro pneumatic servo actuator converts electrical signals to pneumatic motion.
(b) Differentiate between the power sources used in electro pneumatic and electro mechanical servo actuators.

12. Home Assignment and Follow Up	Prepare a brief note on the role of electro mechanical servo actuators in the next lesson on powered control (electro hydraulic servo motors) to establish a bridge between hydraulic and mechanical/electrical actuation systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principle of operation of powered control (electro hydraulic servo motors).	Refer to the master syllabus reference list	LP 34

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min